

LESSON 106 (2.0) GROUND**Lesson Objective**

During this lesson, the applicant will be introduced to the basic elements of principles and aerodynamics of flight. This lesson should serve as an introduction to the elements listed in the lesson contents.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Ask the student if they have any questions or areas of uncertainty from their readings in AFH Chapters 4 or 5.
- Student questions.

References

- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 04 – Principles of Flight
 - Chapter 05 – Aerodynamics of Flight

Instructor Guidance

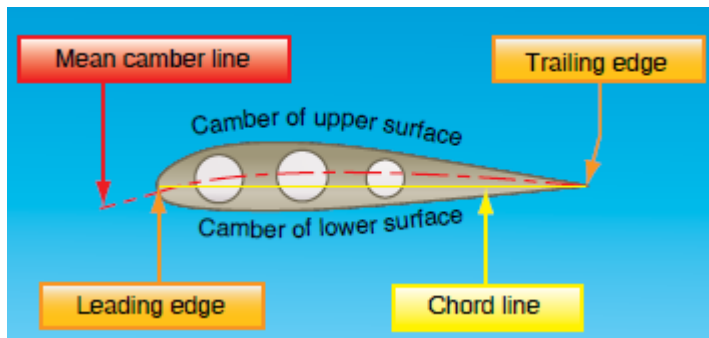
- Keep explanations simple and relatable — avoid overcomplicating topics such as theories of lift, ground effect, and turning tendencies. The goal is comprehension and retention, not aerodynamic mastery.
- Emphasize what the student needs to understand for safe operations. The completion standard calls for a basic understanding — teach to that standard, not beyond it.
- The goal is comprehension and retention, not aerodynamic mastery.

Lesson Content

Structure of the atmosphere

- Composition of the atmosphere (78% Nitrogen, 21% Oxygen, trace gases)
- Layers of the atmosphere (Troposphere, Stratosphere overview)
- Standard atmosphere assumptions (ISA conditions)
- Temperature lapse rate (approx. 2°C per 1,000 ft)
- Pressure variation with altitude
- Air density and its relationship to lift and engine performance
- Density altitude and its operational significance

Airfoil Design

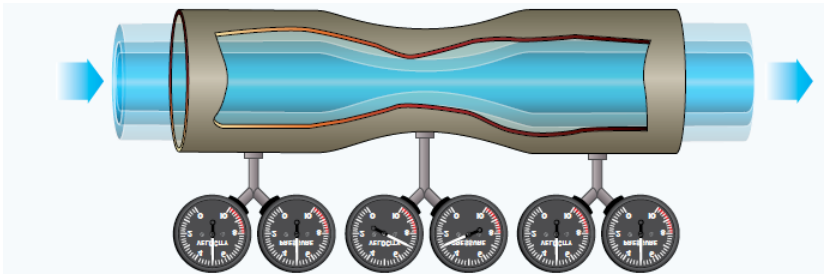


- Leading Edge
 - The front portion of the airfoil that first meets the relative wind.
- Trailing Edge
 - The rear portion of the airfoil where airflow leaves the surface.
- Chord Line
 - An imaginary straight line drawn from the leading edge to the trailing edge.
- Camber
 - The curvature of the airfoil from the chord line. Greater camber generally increases lift at a given angle of attack.
- Mean Camber Line
 - A line halfway between the upper and lower surfaces of the airfoil.
- Thickness
 - The distance between the upper and lower surfaces of the airfoil at its thickest point.
- Angle of Attack (AoA)
 - The angle between the chord line and the relative wind.
- Relative Wind
 - The airflow opposite and parallel to the direction of flight

Theories in the production of lift

- Newtons third law
 - Newton's Third Law states that for every action, there is an equal and opposite reaction.

- A wing at a positive angle of attack deflects airflow downward.
 - The downward deflection of air produces an upward reaction force on the wing.
 - Lift increases as angle of attack increases, up to the critical angle of attack.
 - A stall occurs when the critical angle of attack is exceeded, disrupting airflow and reducing lift
- Bernoulli's Principle
 - As the velocity of a fluid increases, its pressure decreases.
 - Over an airfoil, airflow travels faster over the curved upper surface.
 - Faster airflow above the wing creates lower pressure compared to the higher pressure below the wing.
 - The pressure differential contributes to the upward force known as lift.
 - Lift production results from both pressure differential (Bernoulli) and downward airflow deflection (Newton's Third Law).



Forces acting on the airplane

Lift

- An aerodynamic force produced by the wings.
- Acts perpendicular to the relative wind.
- Created primarily by angle of attack and airflow over the airfoil.
- Must equal weight for straight-and-level, unaccelerated flight.

Weight

- The force of gravity acting on the aircraft.
- Acts vertically downward through the center of gravity (CG).
- Opposes lift.

Thrust

- The forward force produced by the engine/propeller.
- Overcomes drag to create motion through the air.
- Must equal drag for constant airspeed in level flight.

Drag

- The aerodynamic force that opposes motion through the air.
- Acts parallel and opposite to the relative wind.
- Consists of parasite drag (form, skin friction, interference) and induced drag (a result of lift production).
- Induced drag
 - Drag that results directly from the production of lift.
 - Caused by wingtip vortices and the pressure differential between the upper and lower wing surfaces.
 - Increases at higher angles of attack.
 - Greatest at low airspeeds (high AoA conditions such as takeoff, climb, and slow flight).
 - Decreases as airspeed increases.
- Types of Parasite Drag
 - Form Drag
 - Caused by the shape and frontal area of the aircraft.
 - Larger, less aerodynamic shapes create more drag.
 - Skin Friction
 - Caused by friction between the air and the aircraft's surface.
 - Increases with surface roughness and contaminated surfaces.
 - Develops within the boundary layer along the aircraft structure.
 - Interference Drag
 - Caused by airflow disruption where aircraft components meet.
 - Results from turbulent mixing at junctions such as wing-fuselage intersections.
 - Reduced through fairings and smooth aerodynamic design.

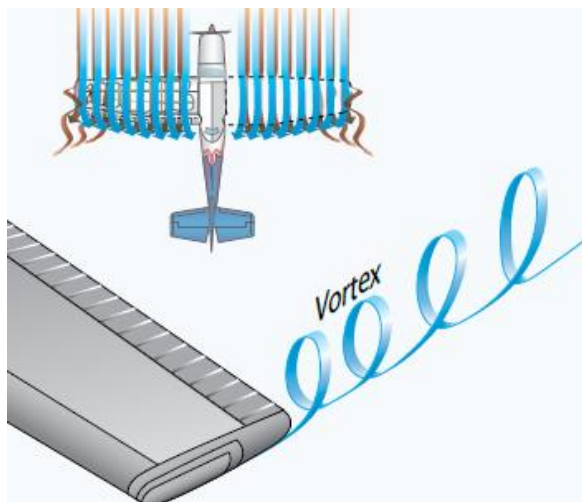
Effect of wing planform

- Aspect Ratio
 - Ratio of wingspan to average chord.
 - High aspect ratio wings reduce induced drag and improve efficiency.
 - Low aspect ratio wings provide greater maneuverability but higher induced drag.
- Wing planform refers to the shape of the wing as viewed from above and influences lift distribution, induced drag, stall behavior, and overall performance.
- Rectangular Wing
 - Produces a relatively uniform lift distribution.
 - Tends to stall at the wing root first, preserving aileron effectiveness.
 - Commonly used on training aircraft due to predictable stall characteristics.
- Tapered Wing
 - Reduces induced drag compared to rectangular wings.
 - More aerodynamically efficient at higher speeds.
 - More susceptible to tip stall if not properly designed.
- Elliptical Wing

- Provides the most efficient lift distribution across the span.
- Produces the least induced drag for a given wingspan.
- More complex and costly to manufacture.

Wingtip vortices

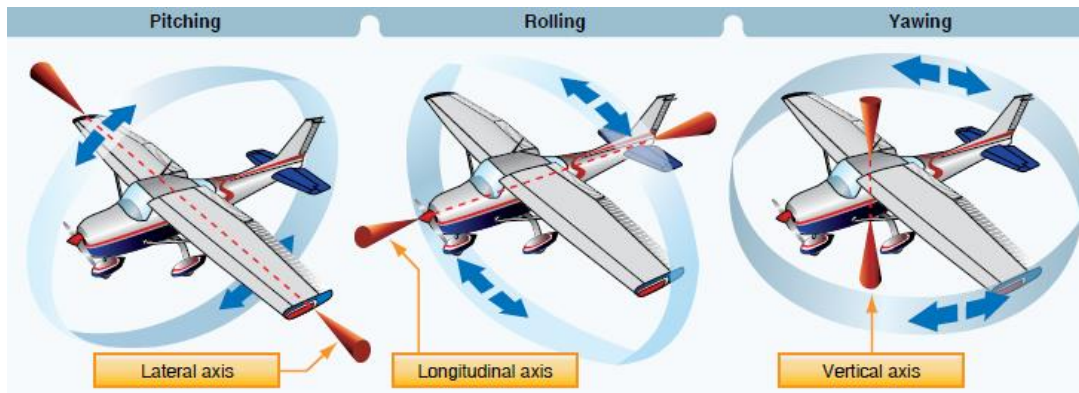
- Wingtip vortices are rotating columns of air formed at the wingtips due to pressure differential between the upper and lower wing surfaces.
- High pressure beneath the wing flows outward and curls upward toward the lower pressure above the wing, creating a spiraling motion.
- These vortices are a direct result of lift production and contribute to induced drag.
- The strength of vortices increases with higher angle of attack, heavier aircraft weight, slower airspeed, and clean wing configurations.
- Wingtip vortices create wake turbulence that can pose hazards to following aircraft.
- Operational avoidance includes staying above the preceding aircraft's flight path and landing beyond its touchdown point.



Ground effect

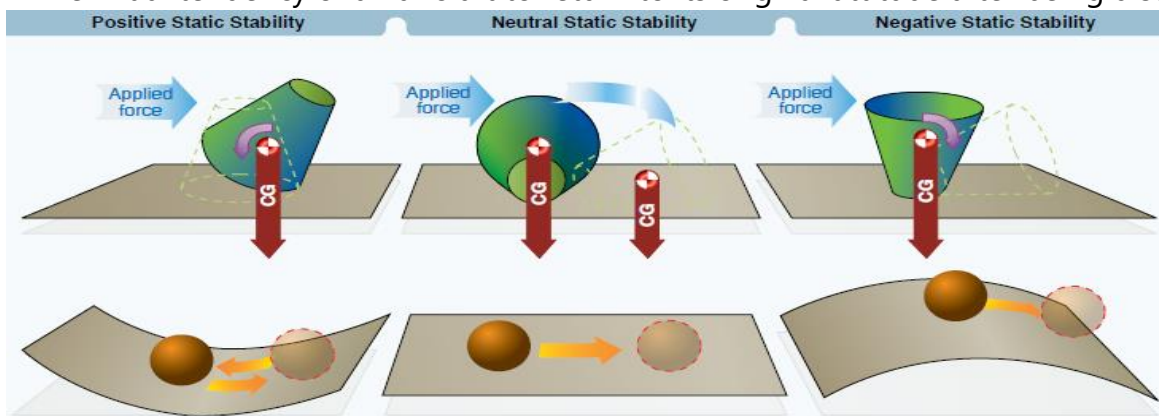
- Ground effect is the reduction in induced drag and increase in lift that occurs when an aircraft operates close to the surface, typically within one wingspan of the ground.
- The proximity of the ground interferes with wingtip vortex formation, reducing downwash and induced drag.
- Because induced drag decreases, the wing becomes more aerodynamically efficient and produces more lift at the same angle of attack.
- During takeoff, ground effect allows the aircraft to become airborne at a lower indicated airspeed, but climb performance may initially be reduced once leaving ground effect.
- During landing, reduced drag and increased lift can cause the aircraft to float if excess airspeed is carried into the flare.
- Ground effect is more noticeable on aircraft with low-mounted wings compared to high-wing configurations.

Axes of an aircraft



Aircraft design characteristics

- Aircraft design characteristics refer to features that influence stability, controllability, performance, and handling qualities. These characteristics determine how the aircraft responds to control inputs and external disturbances in flight.
- Static Stability
 - The initial tendency of an aircraft to return to its original attitude after being disturbed



- Dynamic Stability
 - The response of an aircraft over time after a disturbance, describing how oscillations increase or decrease.
- Longitudinal Stability
 - Stability about the lateral axis (pitch). Primarily influenced by the horizontal stabilizer and center of gravity location.
- Lateral Stability
 - Stability about the longitudinal axis (roll). Enhanced by dihedral, wing position, and keel effect.
- Directional Stability
 - Stability about the vertical axis (yaw). Primarily influenced by the vertical stabilizer.
- Center of Gravity (CG) Location
 - Forward CG increases stability but requires greater control forces.
 - Aft CG decreases stability and may result in reduced stall recovery capability.
- Wing Configuration

- High-wing aircraft tend to exhibit greater lateral stability due to pendulum effect. Low-wing aircraft may provide increased maneuverability.
- Dihedral
 - Upward angle of the wings that contributes to positive lateral stability.
- Tail Configuration
 - Conventional and T-tail designs affect pitch stability and stall behavior.

Basic propeller principles

- A propeller is a rotating airfoil that produces thrust by accelerating air rearward.
- Each propeller blade is shaped like an airfoil and produces lift as it rotates through the air.
- The lift produced by the rotating blades acts forward, creating thrust.
- Blade Pitch
 - The angle between the blade chord line and the plane of rotation. Affects the amount of air accelerated and engine load.
- Fixed-Pitch Propeller
 - Blade angle is set and cannot be adjusted in flight. Engine RPM varies with airspeed and power setting.
- Constant-Speed Propeller
 - Blade angle automatically adjusts to maintain a selected RPM. Allows optimal efficiency across different flight conditions.
- Angle of Attack of the Propeller Blade
 - Determined by blade pitch and relative wind. Changes as airspeed changes.
- Left-Turning Tendencies
 - Torque – reaction force causing roll opposite propeller rotation.
 - P-factor – asymmetric thrust during high angle of attack.
 - Spiraling Slipstream – rotating airflow striking the vertical stabilizer.
 - Gyroscopic Precession – force resulting from rotating mass during pitch changes.
- Propeller efficiency varies with airspeed, blade design, and operating conditions.

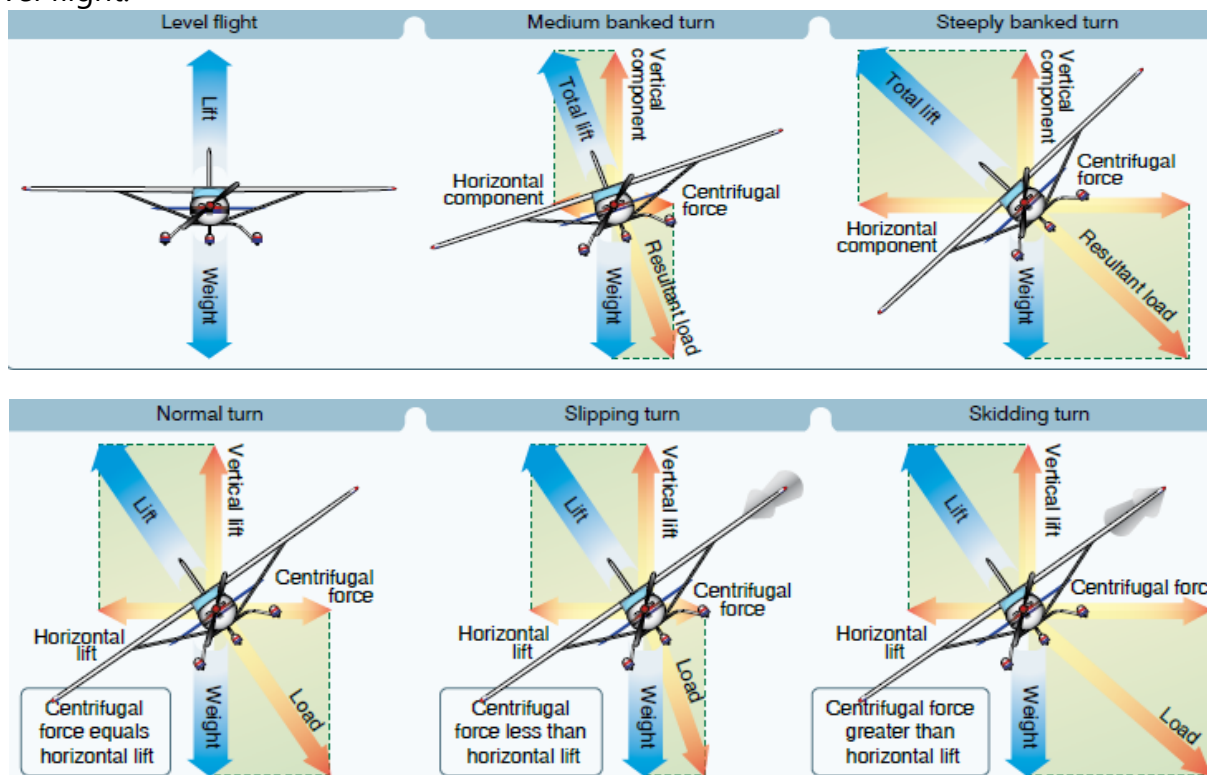
Aerodynamic forces in flight maneuvers

- During flight maneuvers, the relationship between lift, weight, thrust, and drag changes as the aircraft accelerates or changes direction.

In straight-and-level unaccelerated flight, lift equals weight and thrust equals drag.

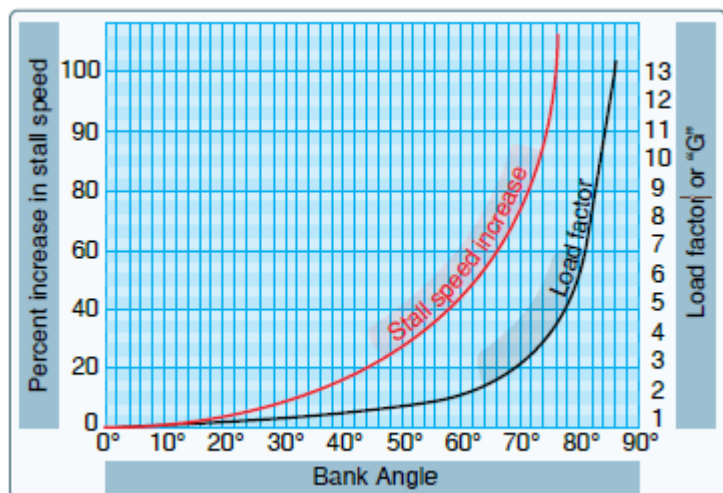
- In a climbing maneuver, lift acts perpendicular to the relative wind, while a component of thrust contributes to overcoming weight. Excess thrust allows the aircraft to gain altitude.
- In a descending maneuver, weight exceeds lift, causing the flight path to angle downward.
- In a coordinated turn, the lift vector tilts. The vertical component of lift supports weight, while the horizontal component provides centripetal force required for turning flight.
- As bank angle increases, total lift must increase to maintain altitude. This requires a higher angle of attack.
- Increasing angle of attack during maneuvering increases induced drag and load factor.

- If sufficient lift is not produced in a turn, the aircraft will descend. If excessive angle of attack is applied, a stall can occur.
- Maneuvering flight results in higher aerodynamic loads on the aircraft compared to straight-and-level flight.



Load factor

- It is expressed in units of G (gravitational force), where 1 G equals the force of weight in straight-and-level, unaccelerated flight.
- In straight-and-level flight, load factor is 1 G.
- In a coordinated level turn, load factor increases as bank angle increases because total lift must increase to maintain altitude.
- As load factor increases, stall speed increases proportionally to the square root of the load factor.
- Example: At 60° of bank, load factor is approximately 2 G, resulting in a stall speed increase of about 41%.
- Positive load factors occur during pull-ups and level turns.
- Negative load factors occur during push-overs and abrupt downward maneuvers.
- Exceeding structural load limits can result in structural damage or failure.
- Understanding load factor is essential for maneuvering flight and respecting aircraft operating limitations.



Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 107
- Assign homework:
 - Review AFH Chapters 4 and 5 to reinforce the aerodynamics and principles of flight covered in today's lesson.
 - Begin memorizing V-speeds from the PIM, as these will be quizzed in upcoming lessons

Completion Standards

This lesson is complete when the applicant can:

1. Demonstrate a basic understanding of the elements of principles and aerodynamics of flight.